Batched subtree queries on Jac GPU

Production CSR Jac • checked LLVM O2 Jac • handwritten checked CUDA

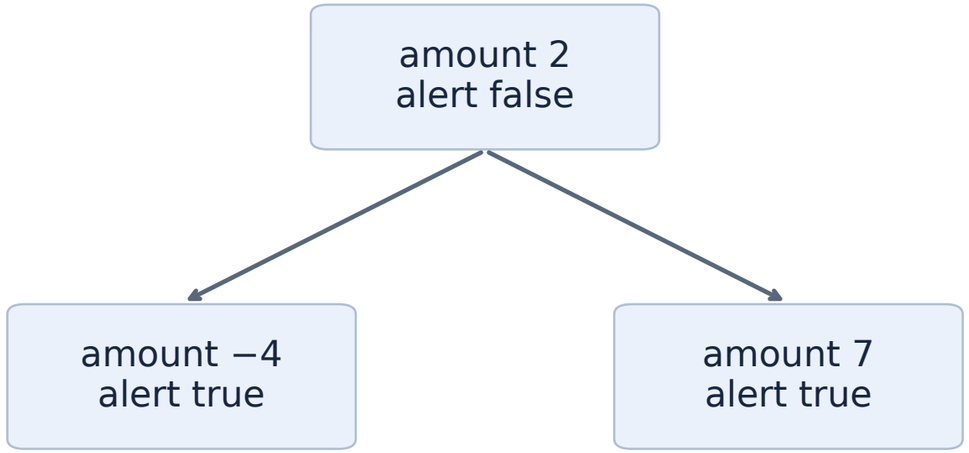
How does general graph traversal compare when every implementation executes the same subtree queries?

17 workloads cover balanced and irregular trees, varied query batches, negative values, leaves and thresholds.

RTX 3090 • shared GPU context • resident actual kernel durations

One query: root, threshold and five results

Traverse the root and every descendant; match amount ≥ threshold.



Query	count / sum / alert / visited
root; threshold 0	2 / 9 / true / 3
left leaf; threshold −4	1 / −4 / true / 1
right leaf; threshold 8	0 / 0 / false / 1

The alert flag combines qualifying nodes. The threshold is retained as the fifth state field. Every query owns its count, sum, flag, visited count and FIFO.

Three implementations share one packed graph

Fixed production pack_walkers discovery order; identical CSR, roots and thresholds.

Group	Compilation and traversal
Original Jac	Production CSR LLVM → existing target-machine codegen
Checked optimized Jac	Same raw LLVM → standard LLVM O2 → target codegen
Handwritten CUDA	Fixed checked per-query CSR FIFO kernel; same predicates

All primary kernels use one thread per query, 128-thread blocks, slot-major FIFO scratch, five outputs and full status checks.

The checked O2 compilation path is experimental. Each arm has private outputs/scratch and shared immutable input pointers.

Full workloads fit a measured VRAM budget

Exact live FIFO capacity is computed over every query.

Factor	Sweep / evidence
Tree shape	Binary, ternary, quaternary, ragged, chain, irregular
Tree size / depth	37…16,383 nodes; depth 5…2,047
Query batches	1…1,024; root, mixed subtree and leaf queries
Complete CPU-checked work	632,829 node visits across 17 workloads
Free VRAM / peak 8-arm plan	370.48 MB / 33.85 MB

Scratch bytes = queries × exact FIFO capacity × 8 per arm. The GPU is shared with Ollama, which held approximately 23,462 MiB.

Three-way result: balanced and ragged trees

Median actual kernel duration in µs; all three columns use one interleaved run.

Workload	Original Jac	Checked Jac + O2	Handwritten
ragged37-mixed33	17.504	18.208	17.568
binary511-root32	241.249	263.233	244.513
binary511-mixed256	173.217	184.513	175.233
binary4095-root16	2,020.425	2,174.922	2,048.041
binary4095-mixed256	2,055.066	2,244.346	2,076.489
binary4095-leaf1024	1.921	1.952	1.920
binary4095-small1024	3.488	3.456	3.392
ternary3280-root16	1,651.336	1,766.296	1,671.896
quaternary4093-mixed128	2,044.729	2,203.402	2,069.994

120 measured kernel samples per cell • 3 warmup rounds • 15 measured rounds × 8 graph nodes • seed 928

Three-way result: chain, irregular and controls

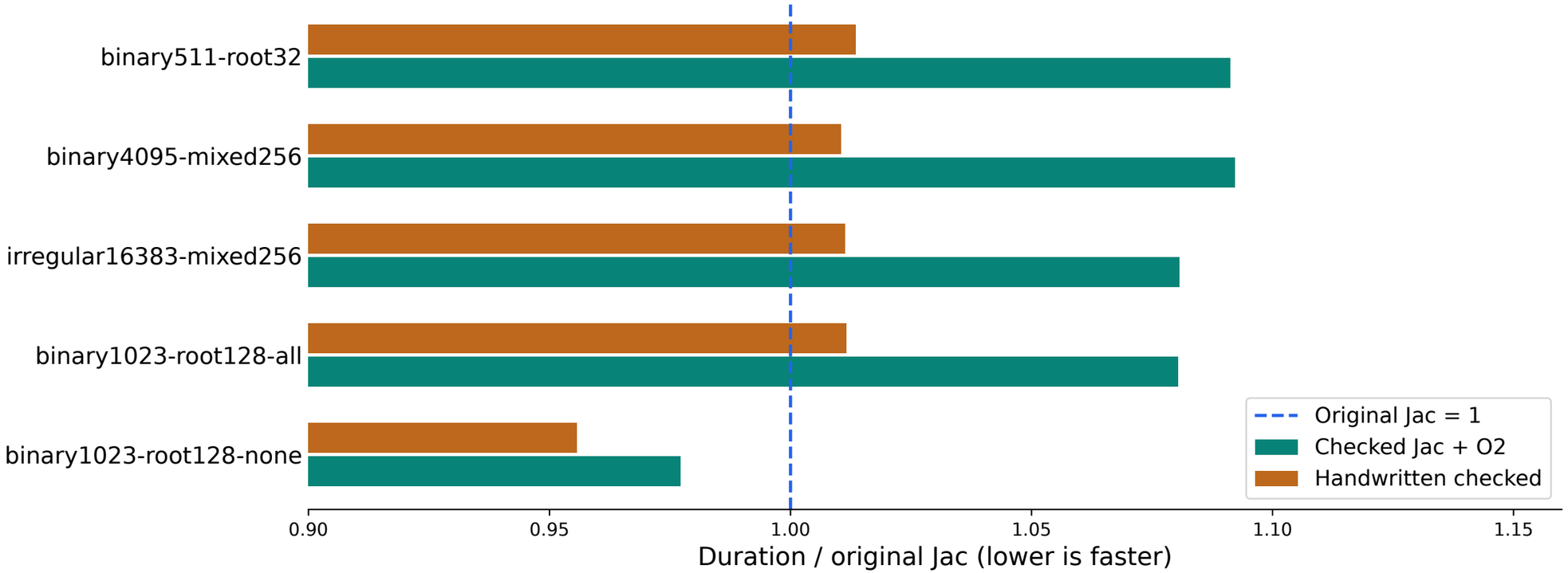
Median actual kernel duration in µs; all three columns use one interleaved run.

Workload	Original Jac	Checked Jac + O2	Handwritten
chain2048-mixed32	748.580	789.028	755.396
irregular4095-mixed128	1,651.208	1,863.513	1,672.072
irregular16383-mixed256	8,271.240	8,938.796	8,365.000
binary1023-root1	403.202	441.122	408.866
binary1023-root32	488.354	524.995	493.634
binary1023-root128	517.443	557.795	523.235
binary1023-root128-all	507.091	547.811	512.963
binary1023-root128-none	469.122	458.386	448.322

120 measured kernel samples per cell • 3 warmup rounds • 15 measured rounds × 8 graph nodes • seed 928

Original CSR Jac closely tracks checked CUDA

LLVM O2 usually adds latency in this sweep; both variants retain full checks.



Irregular 16,383-node batch: 8.271 ms original → 8.939 ms O2 → 8.365 ms handwritten.

This backend already uses SSA state

Single-cursor CSR lowering is inspected independently.

Property	Original Jac	Checked O2	Handwritten
LLVM allocas	0	0	CUDA source
PTX local load / store sites	0 / 0	0 / 0	0 / 0
Driver-JIT local bytes	0	0	0
Driver-JIT registers	38	40	40

FIFO positions, pending/step counters and query states are LLVM phi values. SROA-only emits identical PTX to the production baseline.

O2 changes generated code, including static global-store sites 13 → 8. Hardware attribution for its measured slowdown remains unresolved.

Batching and matching work change latency

Graph hashes and control identity are verified against the saved arrays.

Control on the same 1,023-node tree	Visits	Original Jac (µs)
1 root query, threshold −1001	1,023	403.202
128 root queries, threshold −1001	130,944	507.091
128 root queries, threshold +1001	130,944	469.122

128× the visits takes 1.26× the latency in the all-match batch contrast.

All-match and no-match queries both traverse every node. Their timing contrast includes the count, sum and alert work on qualifying nodes.

Controlled check and FIFO layout experiments

Mixed: 4,095 nodes / 256 queries. Uniform: 1,023 nodes / 128 queries. Times in ms.

LLVM check experiment	Kernel ms
Checked O2	2.261
Bypass overflow	1.859
Bypass validation	1.992
Bypass both	1.597

CUDA scratch layout	Mixed	Uniform root
Slot-major checked	2.093	0.516
Query-major checked	1.503	0.574

Bypass variants use validated finite, bounded tree inputs. Their elapsed times compare complete generated implementations.

Query-major scratch is faster for mixed queries and slower for uniform full-root batches here. Cache/coalescing/occupancy attribution remains open.

Correctness covers valid work and status paths

All five result fields and status are checked after every replay.

Evidence	Result
Serial Jac + independent CPU reference	17 full workloads; 632,829 query-node visits
Direct checked GPU fixtures	17 fixtures × 4 kernels = 68 passing launches
Primary launch name/order/grid/block audit	7,395 verified; 6,120 measured nodes
Supplementary launch audit	19,720 verified; 16,320 measured nodes
Existing subtree CPU/JIT suite	4 tests passed

Fixtures include values above 2^53, negative sums, overflow, private state, malformed CSR, queue/visit limits, cycles and empty roots.

Output/status poisoning happens before the event interval. Error fixtures compare directly with the unmodified production LLVM JIT.

A graph-to-graph comparison gives a closer baseline

The measured task is generic subtree traversal over the same CSR inputs.

Original production CSR Jac is close to the straightforward checked handwritten kernel across this sweep.

Checked LLVM O2 preserves semantics and usually adds latency here. This lowering already has zero local stack storage.

Batch size, qualifying-node work and FIFO layout have measured effects. The hardware causes of remaining differences stay open.

REPORT.md • primary/comparison.csv • raw traces/results • reproducible scripts and source hashes